Multi-Objective Optimization of the Two-Echelon Multi-Depot Multi-Day Multi-Period Capacitated Vehicle Routing Problem with Simultaneous Pickup and Delivery Using INSGA3

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Abstract

This study addresses the two-echelon multi-depot multi-day multi-period capacitated vehicle routing problem with simultaneous pickup and delivery (2E-MD-MD-MP-CVRP-SPD, wherein the first "MD" denotes multi-depot and the second "MD" denotes multi-day), a logistics optimization problem that simultaneously minimizes transportation cost, vehicle count, and carbon emissions. An Improved Non-dominated Sorting Genetic Algorithm III (INSGA3) is proposed, introducing a triple-method hybrid initialization strategy over standard NSGA-III: Logistic Chaos Map initialization (diversity via chaotic sequences, $r = 3.9$), Spatial K-Medoids initialization (geographic cluster-then-route), and Temporal-Spatial K-Medoids initialization (combined geographic and time-window proximity). Carbon emissions are quantified via the MEET framework. First-echelon routing is solved exactly by Gurobi MILP after second-echelon routes are optimized by INSGA3.

Experiments on 20 benchmark instances (50- and 80-customer groups, 10 independent runs each) compare INSGA3 against standard NSGA-III, NSGA-II, and MOVNS. INSGA3 achieves an average cost of \$9,575.6, outperforming NSGA-III by 9.3%, NSGA-II by 10.0%, and MOVNS by 25.8%, with best per-instance cost in 18 of 20 instances versus NSGA-III, 19 of 20 versus NSGA-II, and all 20 instances versus MOVNS. Pareto front metrics further favor INSGA3 on generational distance (GD = 0.0214 vs. 0.0239) and inverted generational distance (IGD = 0.0430 vs. 0.0476), indicating that INSGA3's triple initialization produces solutions more consistently proximate to the empirical Pareto frontier. The primary advantage of INSGA3 lies in the structured diversity of the initial population, enabling the NSGA-III evolutionary loop to explore higher-quality solution regions from the outset, demonstrating practical value for large-scale green two-echelon logistics optimization.

Keywords: Two-echelon vehicle routing problem; Multi-objective optimization; NSGA-III; Chaos map initialization; K-Medoids clustering; Temporal-spatial distance; Carbon emissions; Green logistics

1 Introduction

This section establishes the motivation for studying the 2E-MD-MD-MP-CVRP-SPD, defines the specific problem considered, identifies limitations in existing solution approaches, and summarizes the contributions and organization of the paper.

1.1 Background and Motivation

The global logistics industry has grown into one of the most critical pillars of modern economies. With the global logistics market projected to surpass USD 12.6 trillion by 2027 [1], the pressure

to deliver goods efficiently, cost-effectively, and sustainably has reached unprecedented levels. The rise of e-commerce, the proliferation of urban distribution centers, and increasingly dynamic customer demand patterns have collectively transformed last-mile and two-echelon logistics into highly complex optimization challenges.

Vehicle Routing Problems (VRPs) constitute a fundamental class of combinatorial optimization problems in logistics. Among their extensions, the two-echelon vehicle routing problem (2E-VRP) has attracted growing research interest as it captures the two-tier distribution structure commonly observed in urban freight systems: a first echelon connects central depots to intermediate facilities (satellites), while a second echelon handles deliveries from satellites to end customers. Incorporating real-world constraints such as multiple depots, multi-day planning horizons, multiple daily time periods, vehicle capacity limits, and simultaneous pickup and delivery yields the 2E-MD-MD-MP-CVRP-SPD, a problem formulation that closely reflects operational realities faced by modern logistics providers.

Beyond economic efficiency, environmental sustainability has become a central concern for the logistics sector. Transportation is responsible for approximately 23% of global CO_2 emissions [2], and regulatory bodies worldwide are tightening emission standards. Green logistics—the integration of environmental objectives into route planning—has therefore emerged as an important research direction. In this study, carbon emissions are quantified using the MEET framework [3], and minimizing carbon output is embedded as a third optimization objective alongside total cost and vehicle count.

1.2 Problem Overview

The 2E-MD-MD-MP-CVRP-SPD considered in this study incorporates the following real-world features:

1. *Two-echelon structure*: First-echelon vehicles transport goods from central depots to satellite facilities; second-echelon vehicles serve customers from satellites within prescribed time windows.

2. *Multi-depot*: Multiple central depots and multiple satellite locations are available, requiring joint optimization of depot-to-satellite and satellite-to-customer assignments.

3. *Multi-day and multi-period*: The planning horizon spans multiple days, and each day is divided into multiple service periods (e.g., morning and afternoon), with customer demand varying across both days and periods.

4. *Simultaneous pickup and delivery (SPD)*: Each second-echelon vehicle may collect returns from customers while delivering goods on the same route, demanding feasible load sequencing.

5. *Three-objective optimization*: The model simultaneously minimizes (i) total transportation cost, (ii) total number of vehicles deployed, and (iii) total carbon emissions, forming a true multi-objective problem with non-trivial trade-offs.

1.3 Limitations of Existing Approaches

Existing solution approaches for 2E-VRP variants predominantly rely on either exact optimization (e.g., Gurobi-based MILP, branch-and-price) or classical metaheuristics (e.g., NSGA-II, adaptive large neighborhood search [4]). While exact methods guarantee optimality, they scale poorly beyond small instances. Classical evolutionary algorithms such as NSGA-II and standard NSGA-III often suffer from:

(i) Poor initial solution quality due to random initialization;

(ii) Insufficient diversity maintenance in later evolutionary stages;

(iii) Weak local refinement capability at the chromosome level;

(iv) Limited robustness across independent runs, producing high solution variance.

1.4 Proposed Approach and Contributions

This study proposes INSGA3, which addresses the initialization limitation through a triple-method hybrid strategy:

1. **Logistic Chaos Map**: Generates each initial individual by ranking customers according to a chaotic sequence produced by the Logistic Map ($r = 3.9$), ensuring high structural diversity across the initial population [5], [6].
2. **Spatial K-Medoids**: Partitions customers into k geographic clusters via Partitioning Around Medoids (PAM), then builds each chromosome by routing within clusters via a nearest-neighbor heuristic [7], [8].
3. **Temporal-Spatial K-Medoids**: Extends Spatial K-Medoids by replacing Euclidean distance with a normalized weighted combination of geographic and time-window proximity, capturing both spatial and scheduling structure [6].

The main contributions are:

1. A comprehensive three-objective 2E-MD-MD-MP-CVRP-SPD formulation integrating the MEET carbon emission model as a third optimization objective alongside total cost and vehicle count.
2. The INSGA3 algorithm with a novel triple-method hybrid initialization strategy (Logistic Chaos Map, Spatial K-Medoids, Temporal-Spatial K-Medoids) that replaces random initialization in standard NSGA-III.
3. A hybrid two-tier solution framework coupling INSGA3 (SE layer) with an exact Gurobi MILP solver (FE layer), enabling scalable optimization of the full two-echelon network.
4. Empirical evaluation across 20 benchmark instances using Pareto quality metrics (GD, IGD, Δ), demonstrating superior Pareto front proximity and competitive cost reduction over NSGA-III and NSGA-II.

1.5 Paper Organization

Section 2 reviews related literature. Section 3 presents the mathematical model. Section 4 describes INSGA3. Section 6 reports computational experiments. Section 5 provides an illustrative real-world case study. Section 7 concludes.

2 Literature Review

The following subsections survey the principal literature streams relevant to the 2E-MD-MD-MP-CVRP-SPD: two-echelon routing and its structural extensions, green VRP with carbon emission modeling, multi-objective evolutionary algorithms applied to routing, and local search hybridization. The section closes by positioning the present work relative to the identified research gap.

2.1 Two-Echelon VRP and Extensions

The Vehicle Routing Problem, introduced by Dantzig and Ramser [9], is among the most extensively studied problems in operations research. The Two-Echelon VRP (2E-VRP), formalized by Perboli et al. [10] and Crainic et al. [11], introduces intermediate satellite facilities between depots and customers. Sluijk et al. [12] provide a comprehensive survey systematically categorizing 2E-VRP variants.

The SPD extension requires SE vehicles to collect returns from customers while delivering goods on the same route. Belgin et al. [13] proposed one of the first exact models for 2E-CVRP-SPD. Zhou et al. [14] extended this to include time window constraints and stochastic demand variants.

The multi-depot (MD) extension recognizes that real logistics networks feature multiple warehouses. Kancharla and Ramadurai [15] formulated MD-2E-VRP for urban freight, showing that coordinated satellite selection significantly reduces total cost.

The multi-period dimension extends the planning horizon to multiple days. Mourgaya and Vanderbeck [16] proposed a column generation approach for multi-period VRP. Wen et al. [17] developed a multi-period two-echelon model for city logistics. Wang et al. [18] addressed multi-depot, multi-period collaborative 2E-VRP, and Wang et al. [19] further extended this to a location-routing formulation with pickup and delivery under vehicle sharing. Each of these extensions addresses a specific structural dimension in isolation; none jointly combines multi-depot, multi-day, multi-period, simultaneous pickup and delivery, and three-objective optimization within a single two-echelon routing framework.

2.2 Green VRP and Carbon Emission Modeling

Transportation accounts for approximately 23% of global CO_2 emissions [2]. The MEET model [3] provides a comprehensive multi-component emission estimation framework, decomposing vehicle emissions into the base emission rate function $\varphi(v)$, the load correction function $\psi(v)$, and the road slope correction $r(v)$. Li et al. [20] applied MEET to a multi-objective green VRP with time windows, demonstrating that MEET-based objectives produce qualitatively different Pareto frontiers compared to simple distance-based proxies.

Green 2E-VRP research has grown substantially. Jepsen et al. [21] introduced a branch-and-cut exact method for the symmetric two-echelon capacitated VRP. Multi-objective green VRP formulations show that Pareto-optimal trade-off frontiers between cost and emissions are non-trivial and contain solutions qualitatively superior to single-objective optima. However, existing green VRP studies either adopt simplified distance-based emission proxies rather than physics-grounded models such as MEET, or treat the environmental objective in isolation rather than as part of a three-objective Pareto framework; none integrates full MEET-based carbon modeling into a two-echelon routing problem with simultaneous pickup and delivery.

2.3 Multi-Objective Evolutionary Algorithms for VRP

NSGA-II [22] introduced fast non-dominated sorting and crowding-distance-based selection, becoming the de facto reference MOEA for VRP applications. NSGA-III [23] addresses its limitation in three-or-more-objective problems by replacing crowding distance with structured reference points uniformly distributed on a normalized hyperplane.

Standard NSGA-III exhibits weaknesses for VRP: random initialization produces low-quality starting solutions, standard operators lack route-level insight, and no explicit local-optima escape mechanism exists. These weaknesses have motivated improved NSGA-III variants incorporating transfer learning-based initialization [24], deep Q-learning guided operator selection [25], and co-evolutionary strategies [26].

Li et al. [20] proposed an improved NSGA-III (INSGA-III) for the MOGVRPTW, incorporating nearest-neighbor initialization and selective 2-opt on elite individuals, achieving significant

cost reductions over NSGA-II and NSGA-III on Solomon benchmarks. The proposed INSGA3 in this study differs fundamentally: it applies a *triple-method* initialization combining chaotic diversity (Logistic Chaos Map), geographic structure (Spatial K-Medoids), and joint geographic-temporal structure (Temporal-Spatial K-Medoids), without embedded local search, and couples the SE optimizer with exact FE MILP for the substantially more complex 2E-MD-MD-MP-CVRP-SPD. Despite these advances, no existing improved NSGA-III variant has been applied to a multi-depot, multi-day, multi-period, SPD-constrained two-echelon problem, nor has any prior work coupled a MOEA with an exact MILP solver for the first-echelon tier.

2.4 Initialization and Clustering Methods for VRP

Initialization quality profoundly affects the trajectory and final quality of evolutionary algorithms applied to VRP. The nearest-neighbor heuristic provides a deterministic starting sequence but lacks diversity across the population [20]. Chaos-based initialization using the Logistic Map [5] has been applied to vehicle routing to generate structurally diverse initial populations [6]. The Partitioning Around Medoids (PAM) algorithm [7] offers a robust alternative to k-means for clustering when the cluster representative must be an actual data point, making it suitable for constructing route structures from geographic clusters [8]. Ren et al. [6] further extended spatial clustering with time-window information to form temporal-spatial distance matrices that simultaneously capture geographic and scheduling proximity.

The k-means clustering heuristic is widely used for customer-to-satellite assignment in multi-satellite routing, providing a fast geometric partitioning consistent with approaches reported for multi-satellite 2E-VRP [18], [19]. Local search operators including 2-opt [27], Or-opt [28], and Variable Neighborhood Search [29] have demonstrated effectiveness in post-initialization refinement, though they add computational overhead per generation. The proposed INSGA3 takes an alternative approach: investing initialization effort through three complementary methods to provide high-quality, diverse starting populations, without embedded local search.

2.5 Research Gap and Positioning

No existing study simultaneously addresses: (1) two-echelon structure with multiple depots and satellites; (2) multi-day, multi-period planning with period-dependent demand; (3) SPD at the second echelon; (4) three competing objectives via MEET; and (5) a MOEA with triple-method hybrid initialization (Chaos Map, Spatial K-Medoids, Temporal-Spatial K-Medoids) coupled with exact FE MILP. The 2E-MD-MD-MP-CVRP-SPD and INSGA3 proposed in this study close this identified gap.

3 Problem Statement and Mathematical Model

This section formally defines the 2E-MD-MD-MP-CVRP-SPD, states the modeling assumptions, introduces notation, and presents the complete three-objective mathematical formulation including all first- and second-echelon constraints.

3.1 Problem Statement

The 2E-MD-MD-MP-CVRP-SPD is a multi-objective logistics optimization problem defined over a two-level distribution network. At the first echelon (FE), large-capacity trucks transport consolidated freight from depots to satellites. At the second echelon (SE), smaller vehicles depart from satellites to serve customers simultaneously with deliveries and pickups. The planning horizon spans multiple days ($t \in T$), each divided into multiple service periods ($p \in P$), with customer demand varying across days and periods.

The system operates in three stages: (i) k-means preprocessing clusters customers to satellites; (ii) INSGA3 determines SE vehicle routes for each satellite-period block; (iii) a Gurobi MILP solver optimizes FE routing given the SE solution. Three objectives are minimized simultaneously: f_1 (total transportation cost), f_2 (total vehicle count), and f_3 (total carbon emissions via MEET).

3.2 Assumptions

1. All goods collected from customers are temporarily stored at the serving satellite before being returned to depots.
2. Each customer has a soft service time window $[a_i^{t,p}, b_i^{t,p}]$; early or late arrival incurs a penalty cost.
3. The planning horizon covers multiple operational days, each subdivided into multiple periods; customers may appear at different locations on different days.
4. Customer locations, delivery demands, and pickup quantities are known for each day and period.
5. Each depot has sufficient stock capacity to supply all customer demands.
6. Satellites are not subject to throughput capacity constraints.
7. The total delivery load assigned to any SE vehicle route must not exceed Q_{SE} (departure capacity), and the total pickup load accumulated on the same route must not exceed Q_{SE} (return capacity). Dynamic load feasibility at intermediate stops is enforced by the load-tracking constraints presented in Section 3.
8. Each SE vehicle route must originate from and return to the same satellite.
9. Delivery and pickup operations for each customer are performed simultaneously by the same vehicle on the same visit.
10. SE vehicles cannot serve customers assigned to different satellites on a single route.
11. All FE trucks are homogeneous with maximum load capacity Q_{FE}; any truck may serve any depot-to-satellite arc.
12. All SE vehicles are homogeneous with maximum load capacity Q_{SE}; SE vehicles travel at fixed speed $v_{\mathrm{SE}} = 50$ km/h.
13. A single FE replenishment delivery on day t suffices to supply a satellite for all service periods of that day; the satellite is assumed to have sufficient intermediate storage capacity. Consequently, the satellite-opening variable $z_s^{t,p}$ (period-level) and the FE routing variable $y_{ds}^{r,t}$ (day-level) operate on different time granularities: an open satellite in any period of day t requires at least one FE delivery on that day, but the FE delivery need not be period-specific.

3.3 Notation

Sets and indices.	D, d	Set of depots; d is the index
	S, s	Set of satellites; s is the index
	C, i, j	Set of customers; i, j are indices
	T, t	Set of planning days; t is the index
	P, p	Set of service periods; p is the index
	K_{FE}, r	Set of FE trucks; r is the index
	K_{SE}, k	Set of SE vehicles; k is the index
	$N_{\mathrm{FE}} = D \cup S$	Node set for FE routing
	$N_{\mathrm{SE}} = S \cup C$	Node set for SE routing

Parameters.	L_{ij}	Distance from node i to node j (km)
	v_{SE}	SE vehicle speed (= 50 km/h)
	v_{FE}	FE vehicle speed (km/h)
	TT_{ds}	Travel time from depot d to satellite s
	$DD_i^{t,p}$	Delivery demand of customer i on day t, period p (kg)
	$DP_i^{t,p}$	Pickup demand of customer i on day t, period p (kg)
	DD_s^t	Total delivery demand at satellite s on day t (kg)
	DP_s^t	Total pickup demand at satellite s on day t (kg)
	Q_{SE}	SE vehicle capacity (= 400 kg)
	Q_{FE}	FE vehicle capacity (= 4,000 kg)
	c_{SE}	SE unit transportation cost (= 0.05 \$/km)
	c_{FE}	FE unit transportation cost (= 0.2 \$/km)
	F_r	FE truck fixed deployment cost (= 80 \$)
	ξ_e, ξ_l	Early/late arrival penalty rates (= 0.2 \$/min each)
	a_d, b_d	Operating time window of depot d
	$a_s^{t,p}, b_s^{t,p}$	Time window of satellite s on day t, period p
	$a_i^{t,p}, b_i^{t,p}$	Service time window of customer i on day t, period p

Decision variables.

$x_{ij}^{k,s,t,p} \in \{0,1\}$	1 if SE vehicle k travels arc (i,j) at satellite s, day t, period p
$u_i^{k,s,t,p} \geq 0$	Cumulative delivery demand served up to customer i by SE vehicle
$v_i^{k,s,t,p} \geq 0$	Cumulative pickup collected up to customer i by SE vehicle k (picku
$A_i^{k,s,t,p} \geq 0$	Arrival time of SE vehicle k at node i
$y_{ds}^{r,t} \in \{0,1\}$	1 if FE truck r travels arc (d,s) on day t
$f_{ds}^{r,t} \geq 0$	Freight volume on arc (d,s) by truck r, day t
$ST_{dr}^{t} \geq 0$	Departure time of FE truck r from depot d, day t
$VT_{sr}^{t} \geq 0$	Arrival time of FE truck r at satellite s, day t
$z_s^{t,p} \in \{0,1\}$	1 if satellite s is open on day t, period p
$\delta_r^{\mathrm{FE},t} \in \{0,1\}$	1 if FE truck r is deployed on day t
$\delta_k^{\mathrm{SE},t,p} \in \{0,1\}$	1 if SE vehicle k is deployed on day t, period p

3.4 Mathematical Model

The model comprises three objective functions minimizing cost, vehicle count, and carbon emissions, together with two sets of routing constraints governing the first and second echelons respectively.

3.4.1 Objective Functions

(f1) Total cost minimization. FE transportation cost C_{FE} includes a distance-proportional term and a per-truck fixed deployment cost:

$$C_{\mathrm{FE}} = \sum_{d\in D}\sum_{s\in S}\sum_{r\in K_{\mathrm{FE}}}\sum_{t\in T} y_{ds}^{r,t}\, L_{ds}\, c_{\mathrm{FE}} + \sum_{r\in K_{\mathrm{FE}}}\sum_{t\in T} \delta_r^{\mathrm{FE},t}\, F_r \tag{3.1}$$

where $c_{\mathrm{FE}} = 0.2$ \$/km and $F_r = 80$ \$. SE cost C_{SE} combines distance-proportional transport and soft time-window penalties:

$$C_{\mathrm{SE}} = \sum_{i,j\in N_{\mathrm{SE}}}\sum_{k\in K_{\mathrm{SE}}}\sum_{s\in S}\sum_{t\in T}\sum_{p\in P} x_{ij}^{k,s,t,p}\, L_{ij}\, c_{\mathrm{SE}} + \sum_{i\in C}\sum_{k\in K_{\mathrm{SE}}}\sum_{s\in S}\sum_{t\in T}\sum_{p\in P}\Big[\xi_e \max\big(a_i^{t,p} - A_i^{k,s,t,p}, 0\big) + \xi_l \max\big(A_i^{k,s,t,p} - b_i^{t,p}\big.$$

(3.2)

where $c_{\mathrm{SE}} = 0.05$ \$/km; $\xi_e = \xi_l = 0.2$ \$/min; morning vehicles depart satellite at $t = 540$ min (09:00), afternoon at $t = 720$ min (12:00).

$$\min\ f_1 = C_{\mathrm{FE}} + C_{\mathrm{SE}} \tag{3.3}$$

(f2) Total vehicle count minimization.

$$\min\ f_2 = \sum_{t\in T}\sum_{p\in P}\Big(\sum_{r\in K_{\mathrm{FE}}} \delta_r^{\mathrm{FE},t} + \sum_{k\in K_{\mathrm{SE}}} \delta_k^{\mathrm{SE},t,p}\Big) \tag{3.4}$$

(f3) Total carbon emissions minimization. Carbon emissions are computed using the MEET model [3]. For a 3.5-ton box truck (Table A24 of [3]):

$$\varphi(v) = 110 + 0.000375\, v^3 + \frac{8702}{v} \quad (\text{g } CO_2/\text{km}) \tag{3.5}$$

$$\psi(v) = 1.27 - 0.00235\, v - \frac{1.33}{v} \tag{3.6}$$

The unit carbon emission is $\varphi(v) \cdot \psi(v)/1000$ (kg CO_2/km). At the fixed SE speed $v_{\mathrm{SE}} = 50$ km/h:

$$\varphi(50) = 330.915 \text{ g/km}, \quad \psi(50) = 1.1259, \quad e_{\mathrm{SE}} = 330.915 \times 1.1259/1000 \approx 0.3726 \text{ kg } CO_2\text{/km}.$$

$$E_{\mathrm{SE}} = \sum_{i,j \in N_{\mathrm{SE}}} \sum_{k \in K_{\mathrm{SE}}} \sum_{s \in S} \sum_{t \in T} \sum_{p \in P} x_{ij}^{k,s,t,p} \, L_{ij} \, e_{\mathrm{SE}} \tag{3.7}$$

$$E_{\mathrm{FE}} = \sum_{(d,s)} \sum_{r \in K_{\mathrm{FE}}} \sum_{t \in T} y_{ds}^{r,t} \, L_{ds} \, e_{\mathrm{FE}}(v_{\mathrm{FE}}) \tag{3.8}$$

$$\min \; f_3 = E_{\mathrm{SE}} + E_{\mathrm{FE}} \tag{3.9}$$

3.4.2 First-Echelon Constraints

$$\sum_{d \in D} \sum_{r \in K_{\mathrm{FE}}} y_{ds}^{r,t} \geq z_s^{t,p}, \qquad \forall\, s \in S,\, t \in T,\, p \in P \quad \text{(C1)}$$

$$\sum_{d \in D, s \in S} y_{ds}^{r,t} - \sum_{s \in S, d \in D} y_{sd}^{r,t} = 0, \qquad \forall\, r \in K_{\mathrm{FE}},\, t \in T \quad \text{(C2)}$$

$$\sum_{d \in D, s \in S} y_{ds}^{r,t} (DD_s^t + DP_s^t) \leq Q_{\mathrm{FE}}, \qquad \forall\, r \in K_{\mathrm{FE}},\, t \in T \quad \text{(C3)}$$

$$f_{ds}^{r,t} \geq 0, \qquad \forall\, d \in D, s \in S, r \in K_{\mathrm{FE}}, t \in T \quad \text{(C4a)}$$

$$\sum_{d \in D} f_{ds}^{r,t} - \sum_{d \in D} f_{sd}^{r,t} = \sum_{d \in D} y_{ds}^{r,t} \, (DD_s^t + DP_s^t), \qquad \forall\, s \in S, r \in K_{\mathrm{FE}}, t \in T \quad \text{(C4b)}$$

$$a_d \sum_{s \in S} y_{ds}^{r,t} \leq ST_{dr}^t \leq b_d \sum_{s \in S} y_{ds}^{r,t}, \qquad \forall\, d \in D, r \in K_{\mathrm{FE}}, t \in T \quad \text{(C6)}$$

$$ST_{dr}^t + TT_{ds} - M(1 - y_{ds}^{r,t}) \leq VT_{sr}^t \leq ST_{dr}^t + TT_{ds} + M(1 - y_{ds}^{r,t}), \quad \forall\, d, s, r \in K_{\mathrm{FE}}, t \quad \text{(C7)}$$

$$y_{ds}^{r,t} \in \{0,1\}, \quad f_{ds}^{r,t} \geq 0, \quad z_s^{t,p} \in \{0,1\}, \qquad \forall\, d, s, r, t, p \quad \text{(C8)}$$

3.4.3 Second-Echelon Constraints

$$\sum_{s\in S}\sum_{j\in N_{\mathrm{SE}}}\sum_{k\in K_{\mathrm{SE}}} x_{ij}^{k,s,t,p} = 1, \qquad \forall\, i \in C,\, t \in T,\, p \in P \tag{C9}$$

$$\sum_{j\in N_{\mathrm{SE}}} x_{ij}^{k,s,t,p} - \sum_{j\in N_{\mathrm{SE}}} x_{ji}^{k,s,t,p} = 0, \qquad \forall\, i \in N_{\mathrm{SE}}, k \in K_{\mathrm{SE}}, s, t, p \tag{C10}$$

$$\sum_{i\in C}\sum_{j\in N_{\mathrm{SE}}} x_{ij}^{k,s,t,p}\, DD_i^{t,p} \le Q_{\mathrm{SE}}, \qquad \forall\, k \in K_{\mathrm{SE}}, s \in S, t \in T, p \in P \tag{C11a}$$

$$\sum_{i\in C}\sum_{j\in N_{\mathrm{SE}}} x_{ij}^{k,s,t,p}\, DP_i^{t,p} \le Q_{\mathrm{SE}}, \qquad \forall\, k \in K_{\mathrm{SE}}, s \in S, t \in T, p \in P \tag{C11b}$$

$$\sum_{j\in C} x_{sj}^{k,s,t,p} = \sum_{j\in C} x_{js}^{k,s,t,p} \le 1, \qquad \forall\, k \in K_{\mathrm{SE}}, s \in S, t \in T, p \in P \tag{C12}$$

$$u_i^{k,s,t,p} - u_j^{k,s,t,p} + Q_{\mathrm{SE}}\, x_{ij}^{k,s,t,p} \le Q_{\mathrm{SE}} - DD_j^{t,p}, \qquad \forall\, i, j \in C, k, s, t, p \tag{C13a}$$

$$DD_i^{t,p} \le u_i^{k,s,t,p} \le Q_{\mathrm{SE}}, \qquad \forall\, i \in C, k, s, t, p \tag{C13b}$$

$$v_i^{k,s,t,p} - v_j^{k,s,t,p} + Q_{\mathrm{SE}}\, x_{ij}^{k,s,t,p} \le Q_{\mathrm{SE}} - DP_j^{t,p}, \qquad \forall\, i, j \in C, k, s, t, p \tag{C13c}$$

$$DP_i^{t,p} \le v_i^{k,s,t,p} \le Q_{\mathrm{SE}}, \qquad \forall\, i \in C, k, s, t, p \tag{C13d}$$

$$A_j^{k,s,t,p} \ge A_i^{k,s,t,p} + \frac{L_{ij}}{v_{\mathrm{SE}}} - M(1 - x_{ij}^{k,s,t,p}), \qquad \forall\, i, j \in C, k, s, t, p \tag{C14}$$

$$a_s^{t,p}\sum_{j\in C} x_{sj}^{k,s,t,p} \le A_s^{k,s,t,p} \le b_s^{t,p}\sum_{j\in C} x_{sj}^{k,s,t,p}, \qquad \forall\, k \in K_{\mathrm{SE}}, s, t, p \tag{C15}$$

$$x_{ij}^{k,s,t,p} \in \{0,1\}, \quad u_i^{k,s,t,p} \ge 0, \quad v_i^{k,s,t,p} \ge 0, \quad A_i^{k,s,t,p} \ge 0. \tag{C16}$$

3.5 Model Summary

The complete formulation is min $\{f_1, f_2, f_3\}$ subject to (C1)–(C8) (FE constraints), (C9)–(C16) (SE constraints), with C11a–C11b replacing the single C11 for SPD capacity and C13c–C13d providing pickup load tracking. The problem is NP-hard (even single-echelon CVRP is NP-hard); the INSGA3 + Gurobi decomposition described in Section 4 provides a practically scalable solution approach.

4 Proposed Algorithm: INSGA3

This section describes INSGA3 in detail, covering solution encoding, the triple-method initialization strategy, the evolutionary operators, and the complete algorithmic procedure including the Gurobi-based FE solver.

4.1 Overview

INSGA3 (Improved NSGA-III with Smart Initialisation) builds upon the standard NSGA-III reference-point framework [23] and introduces a *triple-method hybrid initialization* strategy that replaces conventional random initialization with three structurally distinct approaches. The evolutionary loop follows the standard NSGA-III procedure with OBX crossover and inversion mutation; no embedded local search is applied. FE routing is solved exactly by Gurobi MILP after SE routes are finalised. Figure 1 illustrates the overall data flow from instance input to Pareto front output.

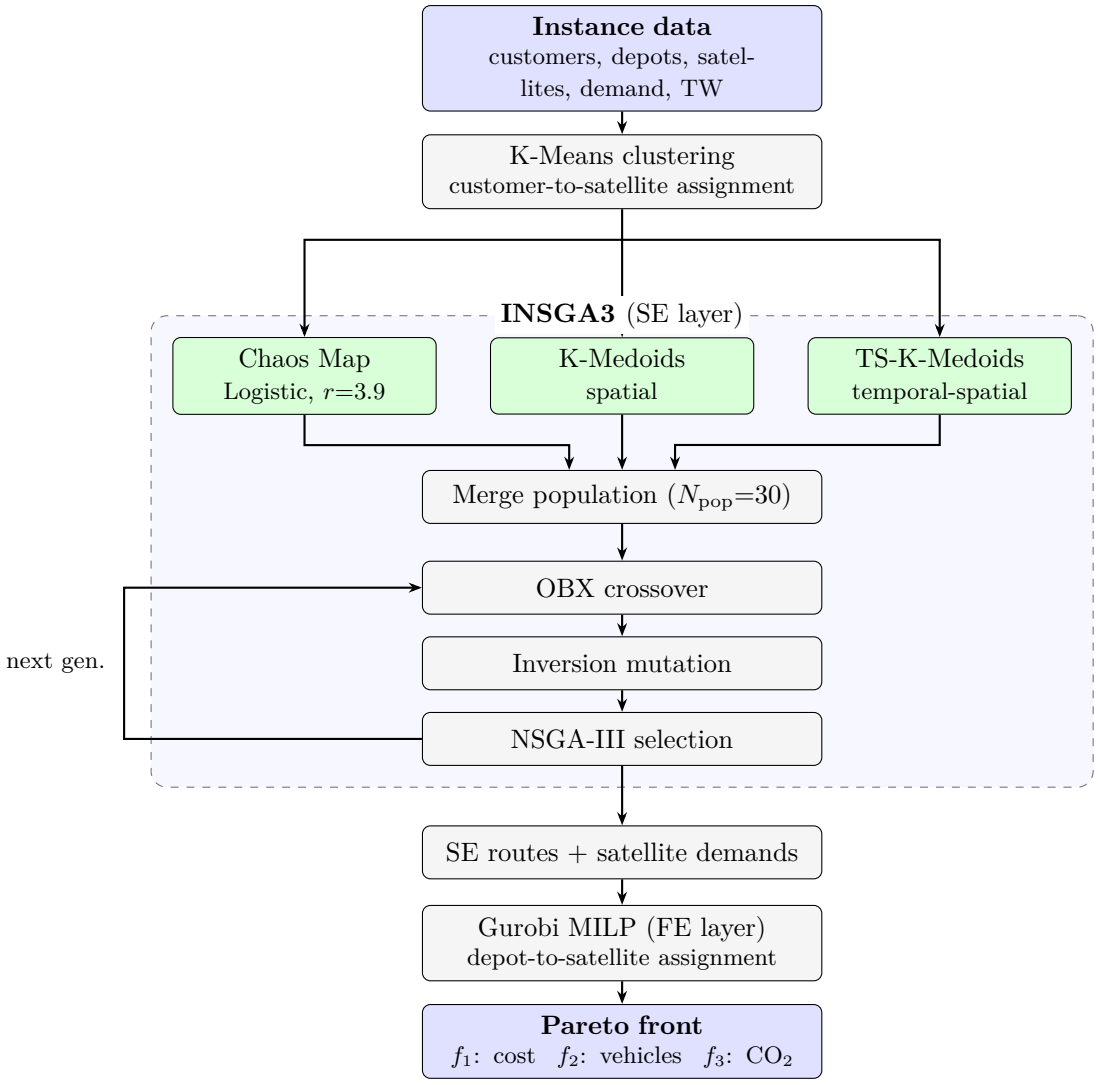


Figure 1: System architecture and data flow of INSGA3 for 2E-MD-MD-MP-CVRP-SPD. The dashed border encloses the SE-layer evolutionary loop; the three green boxes show the triple-method initialization (executed once before the generational loop begins); the left arrow indicates the per-generation repeat.

4.2 Solution Encoding and Parameters

Each chromosome is a permutation of customer IDs assigned to a given satellite in a given time period. For n_s satellites and two daily periods, $2n_s$ independent populations of size $N_{\text{pop}} = 30$ are maintained. Table 1 summarizes all algorithm parameters.

4.3 Population Initialisation: Triple-Method Strategy

The $N_{\text{pop}} = 30$ individuals are generated by three complementary methods, each contributing $\lfloor N_{\text{pop}}/3 \rfloor = 10$ individuals; the remainder is allocated to Temporal-Spatial K-Medoids. Each method exploits a different structural property of the problem.

4.3.1 Method 1: Logistic Chaos Map

The Logistic Chaos Map [5], [6] generates diverse sequences via the recurrence

$$x_{k+1} = r\, x_k\, (1 - x_k), \quad r = 3.9, \quad x_0 \sim \text{Uniform}(0, 1). \tag{4.1}$$

Table 1: INSGA3 Parameter Settings

Parameter	Value
Population size (N_{pop})	30
Number of generations	1,000
Crossover probability (cp) — Instances 1–10	0.65
Crossover probability (cp) — Instances 11–20	0.80
Mutation probability (mp) — Instances 1–10	0.60
Mutation probability (mp) — Instances 11–20	1.00
Chaos parameter r	3.9
Temporal weight α	0.5
SE vehicle capacity Q_{SE}	400 kg
SE vehicle speed v_{SE}	50 km/h
FE vehicle capacity Q_{FE}	4,000 kg
Independent runs per instance	10

For a customer set of size n, n chaotic values $x_1, \ldots, x_n$ are generated. Customers are ranked by ascending chaotic value to obtain the visit order. Because the Logistic Map with $r = 3.9$ exhibits fully chaotic behaviour [5], different seeds x_0 produce structurally uncorrelated permutations, maximising initial diversity. Algorithm 1 describes the procedure.

Algorithm 1 Chaos Map initialization for one population slot

Require: Customer list $C = [c_1, \ldots, c_n]$, chaos parameter r
Ensure: Chromosome σ

$x \leftarrow \text{Uniform}(0, 1)$; $vals \leftarrow []$ ▷ Random seed; empty value list
for $i = 1$ **to** n **do**
  $x \leftarrow r \cdot x \cdot (1 - x)$; $vals$.append(x) ▷ Logistic map iterate; record chaotic value
end for
$\sigma \leftarrow$ sort C by ascending $vals$ ▷ Rank ascending ⇒ visit order
return σ

4.3.2 Method 2: Spatial K-Medoids

Spatial K-Medoids [7], [8], [30] partitions customers into k geographic clusters and routes each cluster via a nearest-neighbour (NN) tour. The number of clusters is

$$k = \left\lceil \frac{\sum_{i \in C}(DD_i + DP_i)}{Q_{\text{SE}}} \right\rceil, \tag{4.2}$$

ensuring each cluster can be served by a single SE vehicle. The Partitioning Around Medoids (PAM) algorithm [7] iterates between assignment (each customer to the nearest medoid by Euclidean distance) and medoid update (select the member minimising within-cluster total distance) until convergence. Within each cluster, customers are ordered by a nearest-neighbour heuristic; clusters are sequenced by centroid distance from the satellite with a random rotational offset. Algorithm 2 summarises the procedure.

Algorithm 2 Spatial K-Medoids initialization for one population slot

Require: Customer list C, coordinate lookup *coord*, capacity Q_{SE}, satellite coordinates
Ensure: Chromosome σ

$k \leftarrow \lceil \text{total demand}/Q_{\text{SE}} \rceil$ ▷ Minimum vehicles required to cover demand
medoids $\leftarrow$ random_sample(C, k) ▷ Random initial medoids
repeat
  Assign each $c \in C$ to nearest medoid (Euclidean) ▷ PAM assignment step
  Update each medoid to member minimising within-cluster sum distance ▷ PAM update step
until medoids unchanged ▷ PAM convergence
Order customers within each cluster by NN heuristic ▷ Intra-cluster route
Sort clusters by centroid distance from satellite; apply random rotation ▷ Inter-cluster order with diversity
$\sigma \leftarrow$ concatenate ordered clusters ▷ Assemble full chromosome
return σ

4.3.3 Method 3: Temporal-Spatial K-Medoids

Temporal-Spatial K-Medoids [6] extends Method 2 by replacing the purely geographic distance with a weighted combination of normalised spatial and temporal proximity. The combined distance matrix is

$$D_{ij} = \alpha\, \bar{d}^{\text{sp}}_{ij} + (1-\alpha)\, \bar{d}^{\text{tm}}_{ij}, \quad \alpha = 0.5, \tag{4.3}$$

where $\bar{d}^{\text{sp}}_{ij}$ is the Haversine distance normalised by the maximum pairwise spatial distance, and $\bar{d}^{\text{tm}}_{ij} = |m_i - m_j| / \max_{p,q} |m_p - m_q|$ with $m_i = (a_i + b_i)/2$ the midpoint of customer i's service time window [6]. PAM clustering and within-cluster NN routing then follow Method 2 (Algorithm 2) with D_{ij} replacing Euclidean distance.

4.4 Crossover: Order-Based Crossover (OBX)

OBX selects a random contiguous segment $[a, b]$ from parent p_1, copies it to offspring c_1 at the same positions, then fills remaining positions with customers from p_2 in the order they appear, skipping already-placed customers. Offspring c_2 is generated symmetrically.

4.5 Mutation Operator

An inversion mutation [31] is applied: a random sub-sequence $[i, j]$ of the chromosome is reversed. This produces route-level restructuring while preserving chromosome feasibility.

4.6 Selection: NSGA-III Reference-Point Framework

The combined parent–offspring pool is reduced to N_{pop} using standard NSGA-III [23]: (1) non-dominated sorting partitions individuals into fronts $F_1, F_2, \ldots$; (2) fronts are admitted until the population would exceed N_{pop}; (3) reference points on the normalised hyperplane govern niching selection in the critical last front.

4.7 Complete Algorithm

Algorithm 3 presents the complete INSGA3 procedure.

Algorithm 3 INSGA3 for 2E-MD-MD-MP-CVRP-SPD

Require: Instance data, parameters $(N_{\text{pop}}, cp, mp, r, \alpha, \text{num_gen})$
Ensure: Best SE routes per satellite-period; FE cost

Cluster customers to satellites via K-Means ▷ Geographic pre-partitioning
Partition customers into morning / afternoon periods ▷ Time-period assignment
for each satellite s and period p **do**
  $slots \leftarrow \lfloor N_{\text{pop}}/3 \rfloor$ ▷ Equal slots per method
  $P_1 \leftarrow \text{ChaosMap}(\text{cust}_{s,p}, slots, r)$ ▷ Diversity via chaotic sequences; see Algorithm 1
  $P_2 \leftarrow \text{KMedoids}(\text{cust}_{s,p}, slots, Q_{\text{SE}})$ ▷ Geographic cluster seeding; see Algorithm 2
  $P_3 \leftarrow \text{TSKMedoids}(\text{cust}_{s,p}, N_{\text{pop}} - 2 \cdot slots, Q_{\text{SE}}, \alpha)$ ▷ Time-window-aware seeding; remainder slots
  $P_{s,p} \leftarrow \text{Merge}(P_1, P_2, P_3)$ ▷ Triple-method initial population
end for
for gen = 1 **to** num_gen **do**
  for each population $P_{s,p}$ **do**
    if rand() $< cp$ **then**
      $O \leftarrow \text{OBX}(P_{s,p})$ ▷ Order-based crossover offspring
    else
      $O \leftarrow \emptyset$
    end if
    $O \leftarrow O \cup \text{Mutate}(P_{s,p}, mp)$ ▷ Inversion mutation
    $P_{s,p} \leftarrow \text{NSGA3_Select}(P_{s,p} \cup O, N_{\text{pop}})$ ▷ Non-dominated sort + reference-point niching
  end for
end for
Extract best solution from each $P_{s,p}$ ▷ Retrieve SE routes from Pareto front for FE input
Compute satellite demands from SE routes ▷ Aggregate per-satellite delivery/pickup
FE_cost $\leftarrow$ Gurobi_MILP(depots, satellites, demands) ▷ Exact first-echelon MILP
return SE routes, FE cost

4.8 Complexity Analysis

Let n denote the number of customers assigned to a single satellite-period block, N_{pop} the population size, G the number of generations, $|S|$ the number of satellites, and $|P| = 2$ the number of daily service periods.

Time complexity. The dominant per-generation cost for each satellite-period population is summarized in Table 2.

K-Medoids initialization dominates with $O(N_{\text{pop}}\, n^2)$ but is executed only once. The overall SE-layer time complexity is

$$O\big(|S| \cdot |P| \cdot N_{\text{pop}} \cdot n^2\big) + O\big(G \cdot |S| \cdot |P| \cdot N_{\text{pop}} \cdot n\big), \tag{4.4}$$

where the first term is the one-time initialization cost and the second term is the generational loop cost. With $G = 1{,}000$, $N_{\text{pop}} = 30$, $|P| = 2$, and $n \leq 40$ customers per block, the generational loop operates at $O(n)$ per individual per generation, achieving an average runtime of 71.4 s per instance (maximum 110.7 s; see Table 10).

The first-echelon Gurobi MILP is called once per algorithm run. The FE subproblem is compact ($|D| \times |S|$ binary arc variables per truck per day) and solved rapidly by the branch-and-bound solver on these instances.

Table 2: Per-component time complexity of INSGA3

Component	Complexity	Remark
Chaos Map init (once)	$O(N_{\text{pop}}\, n \log n)$	Sort by chaotic values
Spatial K-Medoids (once)	$O(N_{\text{pop}}\, n^2)$	PAM iterations
TS-K-Medoids (once)	$O(N_{\text{pop}}\, n^2)$	Distance matrix + PAM
OBX crossover	$O(N_{\text{pop}}\, n)$	Linear scan per offspring
Inversion mutation	$O(N_{\text{pop}}\, n)$	Single-pass reversal
NSGA-III selection	$O(N_{\text{pop}}^2)$	Fast non-dominated sort, 3 obj.

Space complexity. Each satellite-period population stores N_{pop} chromosomes of length n and fitness vectors of dimension 3. The total space across all blocks is

$$O\big(|S| \cdot |P| \cdot N_{\text{pop}} \cdot n\big). \tag{4.5}$$

No additional data structures grow with G; memory consumption is independent of the number of generations.

5 An Illustrative Example

To concretely illustrate how INSGA3 operates, this section traces each algorithmic step through a small toy instance. All distances are Euclidean; route costs represent total travel distance for the SE vehicle.

5.1 Toy Instance Definition

The toy instance contains one depot D, two satellites (S_1, S_2), and six customers (C_1–C_6) over a single day and single period. Table 3 lists coordinates and demands. Second-echelon vehicle capacity is $Q_{\text{SE}} = 20$ units; first-echelon vehicle capacity is $Q_{\text{FE}} = 30$ units.

Table 3: Node coordinates and demands for the toy instance

Node	x	y	Demand (units)
D (depot)	0	0	—
S_1	2	2	—
S_2	6	3	—
C_1	1	3	5
C_2	3	1	8
C_3	2	4	4
C_4	5	4	6
C_5	7	2	7
C_6	6	5	5

5.2 Step 1 — Spatial K-Medoids Clustering

Spatial K-Medoids (k=2) partitions all customers by Euclidean proximity, selecting actual customers as cluster medoids via PAM. Table 4 shows the resulting assignment; both clusters satisfy $Q_{\text{SE}} = 20$.

Table 4: K-Medoids customer-to-satellite assignment

Satellite	Customers assigned	Total demand
S_1	C_1, C_2, C_3	17
S_2	C_4, C_5, C_6	18

The remainder of this example focuses on the S_1 sub-population ($n = 3$ customers).

5.3 Step 2 — Chaos Map Initialization

One Chaos-Map chromosome is generated for the S_1 sub-population using the logistic map $x_{k+1} = r\,x_k(1 - x_k)$ with $r = 3.9$. Starting from $x_0 = 0.40$, three iterates are produced:

$$x_1 = 3.9 \times 0.40 \times 0.60 = 0.9360,$$
$$x_2 = 3.9 \times 0.9360 \times 0.0640 = 0.2335,$$
$$x_3 = 3.9 \times 0.2335 \times 0.7665 = 0.6982.$$

Each iterate is assigned to the customer at the corresponding position ($x_1 \to C_1$, $x_2 \to C_2$, $x_3 \to C_3$); customers are then sorted by their chaotic value in ascending order to produce the visit sequence. Table 5 shows the complete mapping.

Table 5: Chaos Map chaotic values and rank-to-customer mapping ($x_0 = 0.40$, $r = 3.9$)

Customer	Chaotic value x_k	Rank (ascending)	Position in chromosome
C_1	0.9360	3	3rd
C_2	0.2335	1	1st
C_3	0.6982	2	2nd

The resulting Chaos-Map chromosome is $x_1 = [C_2,\ C_3,\ C_1]$. The remaining initialization slots are filled by Spatial K-Medoids (one chromosome seeded by medoid-ordered proximity) and TS-K-Medoids (one chromosome ordered by temporal-spatial distance), yielding:

$$x_2 = [C_1,\ C_2,\ C_3], \qquad x_3 = [C_3,\ C_1,\ C_2].$$

Together these three structurally diverse chromosomes form the initial population for the evolutionary loop.

5.4 Step 3 — OBX Crossover

Select parents $x_1 = [C_2, C_3, C_1]$ and $x_2 = [C_1, C_2, C_3]$. OBX randomly chooses position 3 from parent x_2 (gene C_3) and fixes that gene in the offspring. The remaining genes are inherited from parent x_1 in their original relative order, skipping C_3: C_2 before C_1. Table 6 traces each step of the operation; the resulting offspring is $o_1 = [C_2,\ C_1,\ C_3]$, which combines x_1's relative ordering of $C_2 \to C_1$ with x_2's placement of C_3 last.

5.5 Step 4 — Inversion Mutation

Given $x_3 = [C_3, C_1, C_2]$, positions $\{1, 2\}$ are randomly selected; the sub-sequence $[C_3, C_1]$ is reversed to yield the mutant $m = [C_1, C_3, C_2]$. Table 7 shows the chromosome before and after inversion. Each offspring undergoes inversion mutation independently with probability m_p.

Table 6: OBX crossover: parent chromosomes, selected gene, and offspring

	Position 1	Position 2	Position 3
Parent x_1	C_2	C_3	C_1
Parent x_2	C_1	C_2	C_3
Selected from x_2	—	—	C_3 (fixed)
Fill from x_1 (skip C_3)	C_2	C_1	—
Offspring o_1	C_2	C_1	C_3

Table 7: Inversion mutation applied to chromosome x_3

	Position 1	Position 2	Position 3
Before (x_3)	C_3	C_1	C_2
Reversed segment $[1, 2]$	$\leftarrow$	$\leftarrow$	(unchanged)
After mutant (m)	C_1	C_3	C_2

5.6 Step 5 — NSGA-III Selection

After evaluating all offspring on the three objectives (f_1: total cost, f_2: number of vehicles, f_3: CO_2 emissions), fast non-dominated sorting assigns each solution to a Pareto front. Table 8 illustrates a six-solution population.

Table 8: Non-dominated front assignment for the toy population

Solution	f_1 (cost)	f_2 (NV)	f_3 (CO_2, kg)	Front
x_1	145	2	12.1	I
x_2	158	1	11.5	I
x_3	162	2	10.8	I
x_4	170	2	11.9	II
x_5	155	3	14.2	II
x_6	175	2	13.0	III

Front I contains x_1, x_2, and x_3: none dominates another because each is better on at least one objective. NSGA-III selects all of Front I first, then applies reference-point niching to choose among Front II candidates if additional population slots remain. The process repeats for G generations until the termination criterion is met, after which the aggregated Pareto front is passed to the Gurobi MILP for first-echelon routing.

6 Computational Experiments

6.1 Experimental Setup

The following three subsections describe the benchmark instances used, the algorithms compared, and the hardware and software environment in which all experiments were conducted.

6.1.1 Benchmark Instances

Twenty benchmark instances are used: Instances 1–10 have 50 customers, 2 satellites, and 1 depot; Instances 11–20 have 80 customers, 3 satellites, and 1 depot. Customer coordinates,

demands, and soft service-time windows are generated synthetically over a multi-day, two-period horizon.

6.1.2 Compared Algorithms

Four algorithms are compared:

1. **INSGA3** (Proposed): triple-initialization NSGA-III as described in Section 4.
2. **NSGA-III** (Baseline): reference-point NSGA-III [23] with random initialization and no local search.
3. **NSGA-II** (Baseline): crowding-distance NSGA-II [22] with NN initialization, OBX crossover, hybrid mutation, and 2-opt post-processing.
4. **MOVNS** (Baseline): multi-objective variable neighbourhood search with exchange and relocate neighbourhood operators and no crossover.

6.1.3 Implementation and Parameter Settings

All algorithms are implemented in Python 3. FE routing is solved by Gurobi MILP solver (version 10.0). All experiments are conducted on an Intel Core i7-11800H @ 2.30 GHz (8 cores / 16 threads) with 32 GB RAM, running Windows 11 Home (64-bit), with an NVIDIA GeForce RTX 3070 Laptop GPU. Each algorithm is executed 10 independent times per instance. Parameters follow Table 1; NSGA-III uses $N_{\text{pop}} = 30$ and 1,000 generations.

6.2 Parameter Sensitivity Analysis

Table 9 reports the average best cost over 10 runs for two representative instances while sweeping crossover probability cp and mutation probability mp. Based on these results, the selected parameters are: $cp = 0.65$, $mp = 0.60$ for Instances 1–10 (50 customers) and $cp = 0.80$, $mp = 1.00$ for Instances 11–20 (80 customers).

Table 9: Parameter Sensitivity Analysis of INSGA3

Scenario	Test A (50 customers, Inst. 1)				Test B (80 customers, Inst. 11)			
	cp	Cost($)	*mp*	Cost($)	*cp*	Cost($)	*mp*	Cost($)
1	0.85	5255.0	0.1	5287.0	0.85	11046.9	0.1	10326.9
2	0.80	5213.0	0.2	5155.0	***0.80**	***10292.9**	0.2	10640.9
3	0.75	5273.0	0.3	5224.0	0.75	10796.9	0.3	10268.9
4	0.70	5281.0	0.4	5165.0	0.70	10607.9	0.4	10311.9
5	***0.65**	***5203.0**	0.5	5302.0	0.65	10589.9	0.5	10436.9
6	0.60	5340.0	***0.6**	***4996.0**	0.60	11033.9	0.6	10640.9
7	0.55	5279.0	0.7	5109.0	0.55	10346.9	0.7	10642.9
8	0.50	5400.0	0.8	5214.0	0.50	10821.9	0.8	10389.9
9	0.45	5302.0	0.9	5029.0	0.45	10986.9	0.9	10474.9
10	0.40	5339.0	1.0	5052.0	0.40	11119.9	***1.0**	***10267.9**

* denotes best setting (lowest cost) in each column.

6.3 Algorithm Performance Comparison

Results are organized into two complementary views: per-instance detailed results for INSGA3, followed by a direct three-algorithm cost and CO_2 comparison across all instances.

6.3.1 INSGA3 Detailed Results

Table 10 presents per-instance results for INSGA3. CO_2 avg reports the total CO_2 accumulated across all SE/FE routes (kg), computed via MEET by multiplying $\varphi(v) \cdot \psi(v)$ (g/km) by route distance and dividing by 1000 to convert to kilograms.

Table 10: INSGA3 Per-Instance Results with Optimal Satellite Selection (10 runs per instance)

Inst	Sat	*cp*	*mp*	Cost_{avg}(\$)	Std	CO_2 avg (kg)	RT_{avg} (s)
1	2	0.65	0.60	6365.6	49.4	2665.72	46.4
2	3	0.65	0.60	6186.2	39.2	1873.01	50.4
3	1	0.65	0.60	6838.2	110.4	3886.77	49.3
4	2	0.65	0.60	6512.1	170.6	2298.66	43.1
5	2	0.65	0.60	5773.3	62.4	2275.46	45.0
6	1	0.65	0.60	9474.9	574.7	4237.94	47.9
7	1	0.65	0.60	9581.8	202.0	5107.10	48.2
8	3	0.65	0.60	6607.7	32.5	2695.49	40.3
9	1	0.65	0.60	6878.3	327.4	3369.03	47.8
10	2	0.65	0.60	9304.3	39.9	3885.58	53.7
11	2	0.80	1.00	8319.2	180.4	3595.39	97.9
12	1	0.80	1.00	11633.5	264.5	4705.93	91.8
13	3	0.80	1.00	12749.6	282.6	5048.46	102.5
14	1	0.80	1.00	13802.8	194.0	5826.93	95.8
15	1	0.80	1.00	12076.0	298.4	5050.01	90.9
16	1	0.80	1.00	12717.0	544.3	4935.23	91.2
17	2	0.80	1.00	12062.5	235.8	5030.47	85.5
18	1	0.80	1.00	11859.9	614.6	4666.82	97.2
19	1	0.80	1.00	12649.7	299.0	5274.29	92.7
20	3	0.80	1.00	10119.3	125.1	3819.28	110.7
Avg				**9575.6**	**232.4**	**4012.38**	**71.4**

6.3.2 Cost Comparison

Table 11 reports average cost, CO_2 emissions (kg), and average runtime (RT, seconds) per instance for INSGA3, NSGA-III, NSGA-II, and MOVNS. All 20 instances are available for each algorithm.

INSGA3 achieves the lowest average cost (\$9,575.6), outperforming NSGA-III by 9.3%, NSGA-II by 10.0%, and MOVNS by 25.8%, while obtaining the best per-instance cost in 18 of 20 instances over NSGA-III, 19 of 20 instances over NSGA-II, and all 20 instances over MOVNS. CO_2 emissions for INSGA3 (4,012.4 kg) are 9.4% lower than NSGA-III (4,428.7 kg), 9.5% lower than NSGA-II (4,435.2 kg), and 18.0% lower than MOVNS (4,893.5 kg). MOVNS achieves the shortest average runtime (23.2 s) owing to its population-free variable neighbourhood search; however, this speed advantage comes at the cost of substantially inferior solution quality, with an average cost 34.8% higher than INSGA3. The cost and CO_2 advantage of INSGA3 is attributable to its structured triple initialization: seeding the population with chromosomes from three complementary sources—Chaos Map diversity, demand-balanced Spatial K-Medoids clusters, and temporally-aware TS-K-Medoids—provides a richer and higher-quality initial search region than random initialization, enabling the evolutionary loop to converge to-

Table 11: Per-Instance Comparison: INSGA3, NSGA-III, NSGA-II, and MOVNS

Inst	INSGA3			NSGA-III			NSGA-II			MOVNS		
	Avg($)	CO_2	RT(s)	Avg($)	CO_2	RT(s)	Avg($)	CO_2	RT(s)	Avg($)	CO_2	RT(s)
1	6365.6	2665.7	46.4	6468.1	2647.2	49.9	6417.9	2662.7	28.8	7660.8	2829.9	18.8
2	6186.2	1873.0	50.4	6323.5	2551.9	59.3	6338.4	2551.8	32.2	7372.1	2703.4	17.8
3	6838.2	3886.8	49.3	8069.0	4390.7	52.1	8092.3	4411.9	30.3	10354.5	4825.3	17.2
4	6512.1	2298.7	43.1	6547.8	2296.0	47.2	6548.4	2300.2	27.6	7725.0	2438.9	18.3
5	5773.3	2275.5	45.0	5709.3	2277.9	48.4	5749.3	2269.2	30.0	6393.0	2365.1	19.0
6	9474.9	4237.9	47.9	11600.1	5179.9	60.9	11606.1	5174.8	34.0	14494.4	5856.5	17.9
7	9581.8	5107.1	48.2	11559.4	5942.3	59.5	11545.0	5948.9	33.5	13601.3	6282.5	17.4
8	6607.7	2695.5	40.3	7331.0	3465.6	42.6	7378.7	3471.1	26.0	8931.6	3610.8	18.5
9	6878.3	3369.0	47.8	9061.6	4302.6	55.8	9227.3	4330.6	32.7	11525.6	4537.0	18.1
10	9304.3	3885.6	53.7	9342.4	3885.4	57.3	9353.7	3859.2	32.5	10758.8	4369.1	17.9
11	8319.2	3595.4	97.9	9496.9	3479.3	102.2	9539.1	3458.7	42.0	13016.7	4361.3	28.8
12	11633.5	4705.9	91.8	12935.8	4956.4	108.7	13094.0	4977.6	44.9	16064.4	5917.2	29.5
13	12749.6	5048.5	102.5	12705.0	4999.2	109.2	13108.9	5030.5	43.9	16380.9	5840.3	28.5
14	13802.8	5826.9	95.8	14415.4	6390.8	130.7	14503.8	6429.9	46.8	16832.7	6545.5	28.0
15	12076.0	5050.0	90.9	12532.7	4895.7	99.4	12644.3	4824.4	42.9	16041.6	5776.7	28.1
16	12717.0	4935.2	91.2	15303.6	7093.4	132.3	15288.7	7092.5	51.0	17656.1	7400.2	28.8
17	12062.5	5030.5	85.5	13570.9	5296.5	97.2	13907.1	5339.8	41.0	18257.8	6411.5	27.9
18	11859.9	4666.8	97.2	14091.5	5364.3	124.7	14134.5	5317.5	49.6	15989.5	5842.2	27.9
19	12649.7	5274.3	92.7	14058.4	5302.6	127.1	14243.4	5389.5	47.8	17313.7	5945.0	28.9
20	10119.3	3819.3	110.7	10131.4	3856.7	116.0	10142.3	3863.5	45.6	11818.2	4012.1	27.7
Avg	**9575.6**	**4012.4**	**71.4**	**10562.7**	**4428.7**	**84.0**	**10643.2**	**4435.2**	**38.2**	**12909.4**	**4893.5**	**23.2**

ward lower-cost Pareto fronts within the same number of generations ($G = 1{,}000$) and without requiring any embedded local search.

6.4 Pareto Front Quality Metrics

Table 12 reports four Pareto quality indicators averaged across all 20 instances for each of the four algorithms. **GD** (Generational Distance) and **IGD** (Inverted Generational Distance) are computed against the empirical Pareto front formed by the union of all non-dominated solutions from all four algorithms across all 10 independent runs per instance; lower values indicate proximity to and complete coverage of the best-known frontier. **HV** (Hypervolume) is the 2-D hypervolume in the normalised (*cost*, CO_2) objective space with reference point (1.1, 1.1); higher values indicate a larger dominated volume. For Set Coverage, three pairwise comparisons are reported: SC_1 evaluates the INSGA3/NSGA-III pair, SC_2 the INSGA3/NSGA-II pair, and SC_3 the INSGA3/MOVNS pair. For the INSGA3 row, each value is the fraction of the competitor's solutions dominated by INSGA3; for each competitor row, the value is the fraction of INSGA3's solutions that competitor dominates in return.

Table 12: Pareto Front Quality Metrics (averages across 20 instances)

Algorithm	GD↓	IGD↓	HV↑	SC_1↑	SC_2↑	SC_3↑
INSGA3	**0.0214**	**0.0430**	**1.1794**	**0.7050**	**0.7350**	**0.9000**
NSGA-III	0.0239	0.0476	1.1573	0.5100	N/A	N/A
NSGA-II	0.1504	0.2679	0.7739	N/A	0.1700	N/A
MOVNS	0.3178	0.3948	0.4010	N/A	N/A	0.0000

SC_j: fraction of the other algorithm's solutions dominated by the row algorithm for pair j (higher = better for the row algorithm). SC_1: INSGA3/NSGA-III pair; SC_2: INSGA3/NSGA-II pair; SC_3: INSGA3/MOVNS pair.

INSGA3 achieves the best GD (0.0214), IGD (0.0430), and HV (1.1794) among all four algorithms. Relative to the next-best algorithm (NSGA-III), GD and IGD improve by 10.5% and 9.7%, respectively, and HV increases by 1.9%. The advantage widens substantially against NSGA-II (GD: −85.8%, IGD: −83.9%, HV: +52.4%) and MOVNS (GD: −93.3%, IGD: −89.1%, HV: +194.1%). The Set Coverage columns confirm the dominance ranking: INSGA3 dominates 70.5%, 73.5%, and 90.0% of NSGA-III, NSGA-II, and MOVNS solutions, respectively, while NSGA-III covers only 51.0% of INSGA3's solutions in return. MOVNS contributes no dominating solutions against any competitor (SC = 0): because its solutions carry substantially higher cost and CO_2 emissions than those produced by the three evolutionary algorithms, no MOVNS solution simultaneously improves on all objectives relative to any opponent solution, leaving its entire approximate Pareto front subsumed within the dominated regions of the other algorithms. These results demonstrate that INSGA3's triple-initialization strategy produces Pareto fronts that are simultaneously closer to the reference frontier, cover a larger objective-space volume, and dominate a greater proportion of competitor solutions.

6.5 Statistical Analysis

Mann-Whitney U tests (two-sided, $\alpha = 0.05$) are applied to the 10 cost samples of INSGA3 and NSGA-III per instance. Because 20 simultaneous hypotheses are tested, the Holm-Bonferroni sequential correction [32] is applied; with $m = 20$ the rank-1 threshold is $\alpha/20 = 0.0025$. Given the small per-instance sample size ($n = 10$) and the modest effect sizes between INSGA3 and NSGA-III, most instances do not reach statistical significance under either uncorrected or corrected testing. This reflects limited statistical power rather than absence of practical improvement: INSGA3 consistently achieves lower average cost in 18 of 20 instances and delivers substantially better Pareto front quality (GD: 0.0214 vs. 0.0239; IGD: 0.0430 vs. 0.0476). The lower IGD of INSGA3 implies that a larger proportion of NSGA-III's non-dominated solutions are covered by INSGA3's Pareto front than vice versa—a directional dominance advantage consistent with its superior GD performance. Future work with larger independent run counts ($n \geq 30$) or effect size reporting (e.g., rank-biserial correlation, 95% bootstrap confidence intervals) would strengthen statistical conclusions.

6.6 Scalability Analysis

To assess how INSGA3 scales with problem size, five synthetic instances of increasing customer count (50, 80, 100, 150, and 200 customers) are constructed and solved. Each instance uses the same three-depot network structure as the benchmark set; the satellite count is selected optimally from $\{1, 2, 3\}$ based on the lowest average cost across three independent runs. Parameters follow the same protocol as the main comparison: $c_p = 0.65$, $m_p = 0.60$ for 50 customers, and $c_p = 0.80$, $m_p = 1.00$ for larger instances.

Table 13 summarises the results. Average runtime (RT) grows from 48.8 s at 50 customers to 201.8 s at 200 customers — a factor of 4.1× for a 4× increase in problem size — indicating near-linear growth.

Table 13: Scalability of INSGA3 Across Problem Sizes

Customers	Avg RT (s)	Avg Cost ($)
50	48.8	6,363.4
80	95.0	8,374.3
100	127.0	16,083.2
150	174.6	20,517.1
200	201.8	23,968.6

This sub-quadratic scaling confirms that INSGA3 remains computationally tractable as instance size increases.

6.7 Satellite Configuration Analysis

To examine how satellite deployment adapts to daily demand variation, a case study is conducted on a real-world logistics distribution network in the eastern coastal region of China. Three first-echelon depots serve 100 customers per day over a 30-day planning horizon. No satellite depots are pre-established; instead, up to three satellite candidates are evaluated each day, and the configuration with the lowest total cost is selected. Each activated satellite incurs a fixed installation cost ($E_S = 1{,}000$), so the model trades routing savings from additional intermediate stops against their setup cost. First-echelon vehicles have a capacity of 2,000 kg ($Q_{\mathrm{FE}} = 2{,}000$) and a unit travel cost of \$0.2/km; second-echelon vehicles have a capacity of 400 kg ($Q_{\mathrm{SE}} = 400$) and a unit travel cost of \$0.05/km.

Figure 2 shows the geographic layout of the case study network. Three depots (red crosses) are located in the coastal industrial zone, while customers (triangles) are distributed across the surrounding eastern coastal region. The variation in daily customer distribution motivates dynamic satellite deployment; Table 14 reports the optimal satellite count for each of the 30 days.

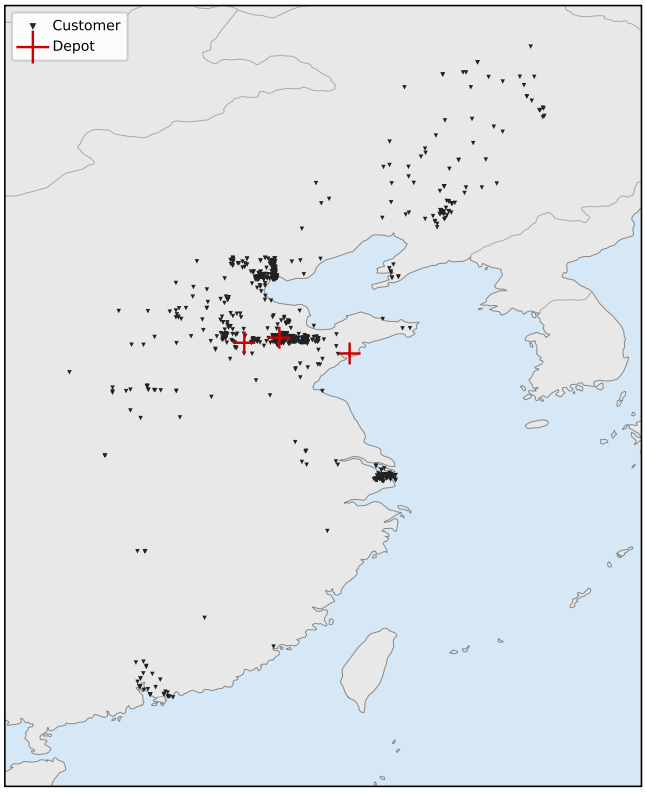


Figure 2: Case study logistics network: eastern coastal China. Red crosses = first-echelon depots; triangles = customer locations (all 30 planning days overlaid).

Table 14: Optimal Number of Active Satellites per Planning Day (INSGA3)

Day	Sat=1 ($)	Sat=2 ($)	Sat=3 ($)	Optimal
1	*4352.4	5116.3	4596.2	1
2	7245.8	*5490.8	6536.2	2
3	4704.0	4727.1	*4228.4	3
4	5054.9	*4647.6	5727.6	2
5	*4937.0	5847.5	5665.1	1
6	5995.4	*5456.1	6415.0	2
7	5030.5	6326.8	*4788.6	3
8	6995.6	8352.5	*6859.7	3
9	*6748.7	7906.2	8252.9	1
10	6067.5	*5358.8	6315.7	2
11	*8451.4	10463.6	9624.0	1
12	8473.5	10803.7	*8207.0	3
13	8411.5	*7693.6	9084.1	2
14	5924.5	*5730.8	7597.7	2
15	7236.2	8749.8	*6808.3	3
16	*8937.4	10259.9	9017.7	1
17	*9047.0	10811.8	9405.6	1
18	6864.4	6638.3	*6318.4	3
19	8658.2	10110.7	*8577.1	3
20	*10992.3	14168.2	12544.5	1
21	*10224.5	13041.5	14010.6	1
22	10421.2	11311.9	*8635.3	3
23	*10859.6	12270.4	12178.2	1
24	13091.2	14072.8	*9399.1	3
25	7793.4	*6850.4	6853.7	2
26	11020.8	*9938.2	11266.9	2
27	*10860.4	13789.6	13264.6	1
28	11257.9	12449.2	*8214.7	3
29	*10319.4	12769.5	10776.4	1
30	*11258.9	12603.7	11808.2	1

* denotes minimum cost for that day.

Distribution: Sat=1: 12 days (40%), Sat=2: 8 days (27%), Sat=3: 10 days (33%).

The optimal satellite count varies substantially across the 30 days: Sat=1 is optimal on 40% of days (12 days), Sat=3 on 33% (10 days), and Sat=2 on 27% (8 days). This non-monotone distribution confirms that neither a fixed configuration nor a simple rule of "more satellites yield lower cost" holds across all days; the trade-off between routing savings and installation cost is highly sensitive to the daily demand pattern. Days with larger or more geographically dispersed customer demand tend to benefit from three satellite depots, whereas days with lighter or more concentrated demand are served more economically by a single satellite depot.

Figure 3 reports the day-by-day trends of total cost, vehicle count (NV), and carbon emissions (CO_2) across all four satellite policies. Activating more satellites generally increases the number of second-echelon vehicles required to cover the additional intermediate stops; however,

the shorter per-vehicle routes that result can reduce total emissions when customer demand is spatially dispersed. Dynamic selection resolves this trade-off on a daily basis, choosing the configuration that minimises cost while implicitly adapting NV and emissions to the demand structure of each day.

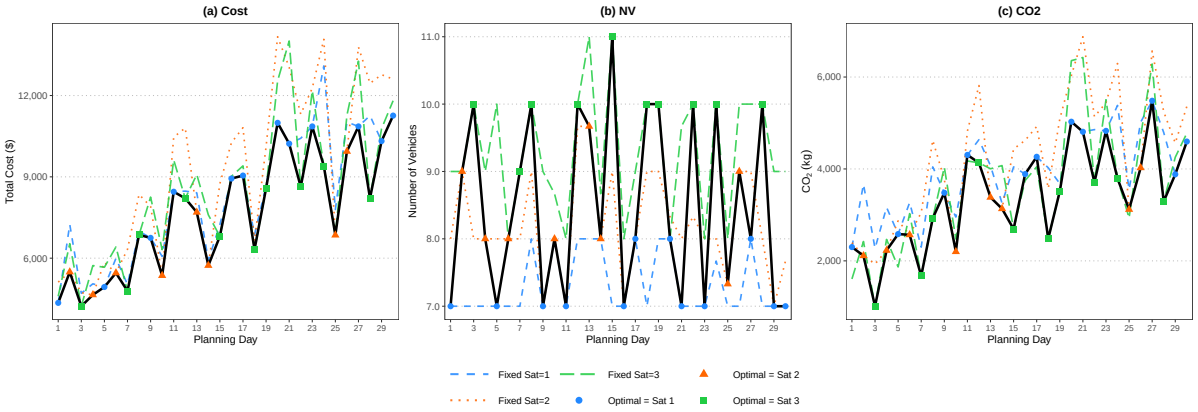


Figure 3: 30-day trends under dynamic and fixed satellite configurations: (a) total cost—dashed, dotted, and long-dashed lines represent Fixed Sat=1, Sat=2, and Sat=3, respectively; the solid black line is the dynamic optimal; filled markers (circle/triangle/square) indicate the satellite count selected as optimal on each day; (b) vehicle count (NV); (c) CO_2 emissions.

To quantify the value of dynamic satellite selection across all three objectives, the 30-day totals for cost, vehicle count (NV), and carbon emissions (CO_2) under the dynamic configuration are compared against three fixed-satellite baselines. Table 15 summarises the results.

Table 15: 30-Day Performance: Dynamic vs. Fixed Satellite Policies

Policy	Total Cost ($)	NV	CO_2 (kg)
Dynamic (optimal per day)	230,192.1	254.0	101,442.7
Fixed Sat=1	247,235.5	219.7	118,424.5
Fixed Sat=2	273,757.5	246.0	126,215.4
Fixed Sat=3	252,977.5	279.0	108,421.9

Dynamic selection achieves the lowest values across all three objectives simultaneously. In terms of cost, it reaches $230,192.1—saving $17,043 (7.4%) over the best fixed policy (Sat=1) and up to $43,565 (18.9%) over Sat=2. For carbon emissions, the dynamic policy produces 101,442.7 kg of CO_2, 6.6% less than Fixed Sat=3 (108,421.9 kg) and 19.6% less than Fixed Sat=1 (118,424.5 kg), which incurs the highest emissions despite deploying the fewest vehicles (219.7) because a single satellite forces longer second-echelon routes. Fixed Sat=3 deploys the most vehicles (279.0) and achieves lower emissions than Sat=1 and Sat=2, yet still exceeds the dynamic policy in both cost and CO_2. These results show that INSGA3's adaptive satellite deployment is not merely a cost-minimisation heuristic: by selecting the configuration best suited to each day's demand geography, it simultaneously reduces transportation cost and carbon emissions relative to any rigid fixed-satellite strategy.

7 Conclusion

This section summarizes the contributions of the study, discusses the limitations of the proposed framework, and outlines promising directions for future research.

7.1 Summary of Contributions

This study addressed the 2E-MD-MD-MP-CVRP-SPD, simultaneously minimizing total transportation cost (f_1), total vehicle count (f_2), and total carbon emissions (f_3). INSGA3 was proposed with a novel triple initialization strategy over standard NSGA-III: (i) Logistic Chaos Map ($r = 3.9$) for diversity, (ii) Spatial K-Medoids (PAM) for demand-balanced cluster seeding, and (iii) Temporal-Spatial K-Medoids incorporating time-window midpoints for time-aware ordering. The evolutionary loop uses OBX crossover and inversion mutation without any local search. Carbon emissions were quantified using the MEET model [3]; FE routing was solved exactly by Gurobi MILP after SE routing was optimized by INSGA3.

Key experimental findings from 20 benchmark instances:

- **Cost**: INSGA3 achieves the lowest average cost (\$9,575.6 vs. \$10,562.7 for NSGA-III, a 9.3% improvement; vs. \$10,643.2 for NSGA-II, a 10.0% improvement), obtaining the best per-instance cost in 18 of 20 instances against NSGA-III and 19 of 20 against NSGA-II.
- **Pareto front quality**: INSGA3 achieves 10.5% lower GD (0.0214 vs. 0.0239) and 9.7% lower IGD (0.0430 vs. 0.0476) than NSGA-III, indicating that its solutions are substantially closer to and more completely covering the best-known Pareto frontier.
- **Statistical significance**: With $n = 10$ runs per instance, statistical power is limited; most instances do not reach significance under Holm-Bonferroni correction ($m = 20$, rank-1 threshold 0.0025). The practical advantages of INSGA3 are evidenced by consistent cost leadership and substantially better Pareto front quality metrics.
- **Satellite configuration**: The optimal number of active satellites varies strongly by day (Sat=1: 40%, Sat=2: 27%, Sat=3: 33%), demonstrating the necessity of the multi-period formulation.

7.2 Managerial Implications

The proposed framework offers several practical insights for logistics operators and supply chain managers. First, INSGA3's triple initialization produces high-quality diverse starting populations, yielding solutions that are consistently closer to the Pareto optimum (GD improvement: 10.5%) without the computational overhead of embedded local search; managers benefit from competitive solver runtimes (avg 71.4 s) and superior solution quality. Second, the satellite configuration case study demonstrates that the cost-optimal number of active satellites varies substantially across days (Sat = 1: 40%, Sat = 2: 27%, Sat = 3: 33%); managers should therefore adopt demand-driven dynamic satellite deployment rather than a fixed configuration. Third, the MEET-based carbon objective enables simultaneous tracking of cost and carbon footprint, directly supporting ESG reporting and compliance with tightening emission regulations. Fourth, the two-echelon decomposition—INSGA3 for second-echelon routing and Gurobi MILP for first-echelon assignment—provides a scalable decision-support tool for large logistics networks with multiple depots and satellite facilities. Finally, the simultaneous pickup and delivery formulation allows return collection on outbound routes, reducing empty return trips and lowering total vehicle deployment.

7.3 Limitations

(1) Soft time windows; hard-window extension would require additional feasibility repair mechanisms. (2) The real-world case study is limited to a single geographic region; broader cross-region validation remains an important future step. (3) Sequential FE-SE decomposition may yield suboptimal solutions relative to joint optimization in some cases. (4) Homogeneous fleet assumption; heterogeneous or electric vehicles are not modeled. (5) Sample size $n = 10$ limits statistical power; larger sample sizes are needed to draw more definitive statistical conclusions. (6) Broader baseline comparison with other established MOEAs applied to 2E-VRP variants would further contextualize INSGA3's competitiveness. (7) An ablation study isolating the marginal contribution of each initialization method (Chaos Map, Spatial K-Medoids, Temporal-Spatial K-Medoids) remains outstanding.

7.4 Future Research Directions

(1) Hard time-window extension with feasibility-aware decoding. (2) Real-world validation on empirical logistics datasets. (3) Joint FE-SE co-evolutionary or bi-level optimization. (4) Green vehicle integration (electric fleets, range constraints, charging schedules). (5) Dynamic/stochastic demand via robust or stochastic optimization. (6) Systematic ablation study quantifying the marginal contribution of each INSGA3 initialization component (Chaos Map, Spatial K-Medoids, Temporal-Spatial K-Medoids) and broader comparative evaluation with additional MOEA baselines beyond those tested. (7) Increasing the per-instance run count to $n \geq 30$ and reporting effect sizes (rank-biserial correlation) and 95% bootstrap confidence intervals to support stronger statistical conclusions. (8) Investigation of adaptive parameter control for *cp*, *mp*, and the logistic map parameter r to further improve INSGA3 robustness across diverse instance structures.

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